Exercise 3.2

1) Let $f(n)$ and $g(n)$ be any two asymptotically non negative functions. By definition of asymptotically non negative, there exists n_0 such that for all $n \geq n_0$, $f(n) \geq 0$ and $g(n) \geq 0$. We fix any $n \geq n_0$. Since both $f(n)$ and $g(n)$ are now non negative, $h(n) = \max\{f(n), g(n)\}$ is non negative. ~~By~~ defi Also:-

$$2h(n) \geq f(n) + g(n)$$

because $h(n) \geq f(n)$ and $h(n) \geq g(n)$. Hence

$$h(n) \geq \frac{1}{2}\left(f(n) + g(n)\right)$$

so $c_1 = 1/2$. Also

$$h(n) \leq f(n) + g(n)$$

so $c_2 = 1$. Since

$$0 \leq c_1\left(f(n) + g(n)\right) \leq h(n) \leq c_2\left(f(n) + g(n)\right)$$

for all $n \geq n_0$, $h(n) = \Theta\left(f(n) + g(n)\right)$

2) Because O-notation represents an upper-bound, not a lower bound, which the phrasing 'at least' implies.

~~3) $2^{n+1} = O(2^n)$~~

3) $2^{n+1} = 2 \times 2^n$ so if we let $c = 2$, then $0 \leq 2^{n+1} \leq 2 \times 2^n$ $\forall n \geq n_0$. Hence $2^{n+1} = O(2^n)$ by definition of O-notation.

Suppose $2^{2n} = O(2^n)$. Then by definition of O-notation, $\exists$ constants c and n_0 such that $\forall n \geq n_0$, $0 \leq 2^{2n} \leq c2^n$.

Since $2^{2n} \leq c2^n$, $c \geq 2^{2n}/2^n$

$c \geq 2^n$

However 2^n grows arbitrarily large as n grows while c is supposed to be a constant and hence a contradiction, therefore our supposition is false and $2^{2n} \neq O(2^n)$.

4) Let $f(n)$ and $g(n)$ be any two functions.

<u>Proof of '$f(n) = \Theta(g(n)) \Rightarrow f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$':-</u>

Suppose $f(n) = \Theta(g(n))$

Then by definition of Θ-notation, $\exists$ constants c_1, c_2 and n_0 such that $\forall\ n \geq n_0$,
$$0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n).$$

Since $0 \leq f(n) \leq c_2 g(n)$, $f(n) = O(g(n))$ by definition of O-notation.

And since $0 \leq c_1 g(n) \leq f(n)$, $f(n) = \Omega(g(n))$ by definition of Ω-notation.

<u>Proof of '$f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$ $\Rightarrow f(n) = \Theta(g(n))$'</u>

Suppose $f(n) = O(g(n))$ and $f(n) = \Omega(g(n))$.

Then by definition of O-notation, $\exists$ constants c_1 and n_1 such that $\forall\ n \geq n_1$, $0 \leq f(n) \leq c_1 g(n)$.

Also by definition of Ω-notation, $\exists$ constants c_2 and n_2 such that $\forall\ n \geq n_2$, $0 \leq c_2 g(n) \leq f(n)$.

Let n_0 be the larger number b/w n_1 and n_2, then $\forall n \geq n_0$,
$$0 \leq c_2 g(n) \leq f(n) \leq c_1 g(n).$$

Hence, by definition of Θ notation, $f(n) = \Theta(g(n))$

5) Proof of 'only if'

Let the running time $T(n) = \Theta(g(n))$. Since $T(n)$ is the running time for every input ~~n~~, it covers both the worst & best case. ~~Since~~ $T(n) = O(g(n))$ in the worst case and $T(n) = \Omega(g(n))$ in the best case (by theorem 3.1).

Proof of 'if'

Let ~~the~~ $T_w(n) = O(g(n))$ and $T_b(n) = \Omega(g(n))$, where $T_b(n)$ is the best case running time and $T_w(n)$ the worst. Then

By definition of O and Ω notations, $\exists$ constants c_1, c_2, n_1, n_2, such that:-

$$0 \leq c_1 g(n) \leq T_b(n) \quad \forall n \geq n_1$$
$$0 \leq T_w(n) \leq c_2 g(n) \quad \forall n \geq n_2$$

Let $n_0 = \max\{n_1, n_2\}$. Then:-

$$0 \leq c_1 g(n) \leq T_b(n) \leq T(n) \leq T_w(n) \leq c_2 g(n)$$

$\forall n \geq n_0$. ~~Since $T_b(n)$~~ Hence by definition of Θ notation, $T(n) = \Theta(g(n))$ ~~(where $T(n)$ is the running time for all inputs of size n)~~.

the set

6) Suppose $o(g(n)) \cap \omega(g(n))$ is not null. Let $f(n) \in o(g(n)) \cap \omega(g(n))$. By definition of $\cap$, $f(n) \in o(g(n))$ and $f(n) \in \omega(g(n))$.

~~By definition of o-notation and ω-notation, ∃ constants $c > 0$, n_1 and n_2 such that for any constant $c > 0$, ∃ constants n_1 and n_2 such that~~

By definition of o-notation and ~~ω-notation~~ ω-notation, ∀ constants $c > 0$, ∃ constants $n_1, n_2 > 0$ such that:-

$$0 \leq f(n) < cg(n) \quad \forall\ n \geq n_1$$
$$0 \leq cg(n) < f(n) \quad \forall\ n \geq n_2$$

Let $n_0 = \max\{n_1, n_2\}$. Then the above inequalities are both true for all $n \geq n_0$, and so $cg(n) < f(n) < cg(n)$ $\forall\ n \geq n_0$, which is a contradiction and so our supposition is false i.e. the set $o(g(n)) \cap \omega(g(n))$ is empty.

7) $\Omega(g(n,m)) = f(n,m) :- \exists\ c, n_0, m_0 \in R^+$ such that $0 \leq cg(n,m) \leq f(n,m)$ for all $n \geq n_0$ or $m \geq m_0$

8) $\Theta(g(n,m)) = f(n,m) :- \exists\ c_1, c_2, n_0, m_0 \in R^+$ such that $0 \leq c_1 g(n,m) \leq f(n,m) \leq c_2 g(n,m)$ for all $n \geq n_0$ or $m \geq m_0$